

高超声速飞行器准平衡滑翔自适应制导

Quasi-equilibrium glide adaptive guidance for hypersonic vehicles

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The entry-glide guidance strategy for hypersonic vehicles that can satisfy both terminal and path constraints is investigated in this paper. We propose a quasi-equilibrium glide adaptive guidance methodology based on the quasi-equilibrium glide condition (QEGC), which innovatively utilizes the quasi-equilibrium glide phenomenon in lifting entry. With the aid of QEGC, both range and terminal velocity can be predicted analytically with high precision. The path constraints are converted into angle of attack constraints, which has been difficult to realize by using traditional predictive guidance methods. The algorithm is independent of the standard trajectory. All the guidance commands, including the bank angle and the angle of attack, are calculated analytically in real time, which endows the algorithm with sufficient adaptability. The results of a CAV-H vehicle guidance test show that the algorithm leads the vehicle along a quasi-equilibrium glide trajectory satisfying both the terminal and path constraints and has sufficient flexibility for occasional mission changes. Furthermore, the robustness of the guidance algorithm under disturbances is validated through a Monte Carlo simulation.

hypersonic vehicles, entry guidance, quasi-equilibrium glide, adaptive guidance, analytical predictor-corrector (APC)

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1 Introduction

Since the 1970s, much interest has been generated around entry guidance methods for high lift-to-drag ratio vehicles with respect to the requirements for developing the space shuttle [1]. Since the end of the 20th century, Mars exploration plans have further promoted the development of entry guidance methods because such plans face a similar atmospheric entry problem [2–6]. Along with the development of 2nd generation reusable launch vehicles (RLVs) such as X-33 and X-37, the entry guidance problem has recently become a popular research topic in the astronautic and aeronautic fields [7–12]. The USA carried out its first successful test of X-37B in 2010. At the end of the last century, investigations for a hypersonic near-space vehicle were ini-

tiated and sponsored by USA. These investigations encouraged new developments in hypersonic glide guidance. In 2010, USA carried out a flight test for their Hypersonic Technology Vehicle 2 (HTV-2), which is considered as an expansion of common aero vehicle (CAV).

The requirement of advanced vehicle entry guidance has promoted the development of many guidance methods. These methods can be divided into two classes: standard trajectory guidance methods and predictive guidance methods. In a standard trajectory guidance algorithm, the standard trajectory must be determined first, then a trajectory (or profile) tracking law is employed. Standard trajectory guidance is mainly realized in two forms: full state following guidance [7] and profile following guidance [1]. Among them, the drag-velocity (D-V) profile tracking guidance law has been successfully used in shuttle entry guidance. The standard trajectory guidance law only accounts for predictable possibilities and lacks the ability to tackle unforeseen

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circumstances during a flight. A class of predictor-corrector guidance algorithms have attracted increasing attention because they offer a guidance method that is independent of the standard trajectory. The common features of these predictor-corrector guidance algorithms are the analytical or numerical propagations of the trajectory to predict the final condition and the correctional steps to adjust the design parameters in order to eliminate the errors in meeting the terminal constraints. Their main advantage is the elimination of the need for a pre-stored reference trajectory, which may differ greatly from the corresponding current state. Based on differences in their predicting algorithms, predictor-corrector guidance methods can be classified into two groups: analytical predictor-corrector (APC) guidance methods and numerical predictor-corrector (NPC) guidance methods. The investigation of APC guidance methods dates back to the 1980s [2], and the methods have mainly been developed for Mars exploration [2–5]. Because the analytical closed-form solution is usually obtained after considerable simplification, the precision of APC guidance methods is low from a traditional perspective. NPC guidance algorithms have attracted even more attention by taking the advantage of modern on-board computational capability [6, 9–12], and efforts are being made to solve the complicated path constraints problem [11, 12]. However, the NPC method requires the complete integration of the remaining trajectory at each guidance cycle, and this requires excessive processing power, making the on-board implementation of this method for actual mission unfeasible. The limitations of the existing guidance and control algorithms prompted NASA's Marshall Space Flight Center to launch the Advanced Guidance and Control (AG&C) Project in 1999. Among all the advanced guidance approaches developed under the AG&C Project, the quasi-equilibrium glide guidance method proposed by Shen and Lu [8] passed all the tests implemented by the Marshall Space Flight Center [13]. The quasi-equilibrium glide guidance method employs the quasi-equilibrium phenomenon during lifting entry to generate trajectory on-board and then tracks the generated trajectory. This quasi-equilibrium glide guidance method depends on the standard trajectory, which is generated online and may still be regarded as the standard trajectory guidance.

In order to reasonably take the advantage of quasi-equilibrium glide guidance as well as eliminate the dependence on the standard trajectory, a novel quasi-equilibrium glide adaptive guidance algorithm based on the predictor-corrector principle is proposed herein for hypersonic vehicles with a high lift-to-drag ratio. Considering the on-board computation constraint, the APC method is preferable. In this method, the trajectory must be converted into a special form to obtain the analytical closed-form solution. An APC with a zero angle of attack assumption for a kinetic warhead was proposed in ref. [14]; however, it is not suitable for glide guidance of vehicles with high lift-to-drag ratios. In this

paper, the quasi-equilibrium glide condition is regarded as the special trajectory form to be converted, based on which the trajectory can be predicted analytically. The use of the quasi-equilibrium glide condition makes the generated trajectory smooth, whereas the special trajectory form of the quasi-equilibrium glide enables accurate analytical prediction.

2 Mathematical model for guidance problem

2.1 Formulation of problem

The three degree-of-freedom (3DOF) point-mass dynamics of a vehicle over a spherical rotating earth are adopted [15]:

$$\dot{V} = -\frac{C_D \rho V^2 S}{2m} - \frac{\mu_M}{r^2} \sin \theta - w_e^2 r (\cos \varphi \sin \varphi \cos \sigma \cos \theta - \cos^2 \varphi \sin \theta), \quad (1)$$

$$\begin{aligned} \dot{\theta} = & \frac{C_L \rho V S}{2m} \cos \nu - \frac{\mu_M}{r^2} \frac{\cos \theta}{V} \\ & + \frac{w_e^2 r}{V} (\cos \varphi \sin \varphi \cos \sigma \sin \theta + \cos^2 \varphi \cos \theta) \\ & - 2w_e \cos \varphi \sin \sigma + \frac{V \cos \theta}{r}, \end{aligned} \quad (2)$$

$$\begin{aligned} \dot{\sigma} = & -\frac{C_D \rho V S}{2m \cos \theta} \sin \nu + w_e^2 r \frac{\cos \varphi \sin \varphi \sin \sigma}{V \cos \theta} \\ & + \frac{2w_e}{\cos \theta} (\cos \varphi \cos \sigma \sin \theta - \sin \varphi \cos \theta) \\ & + \frac{V \tan \varphi \cos^2 \theta \sin \sigma}{r \cos \theta}, \end{aligned} \quad (3)$$

$$\dot{r} = V \sin \theta, \quad (4)$$

$$\dot{\lambda} = -V \cos \theta \sin \sigma / (r \cos \varphi), \quad (5)$$

$$\dot{\varphi} = V \cos \theta \cos \sigma / r, \quad (6)$$

where V is the earth-relative velocity; θ is the flight path angle; σ is the velocity azimuth angle measured from the north in a clockwise direction; r is the radial distance from the center of the earth to the vehicle; λ and φ are the longitude and latitude, respectively. ρ is the atmospheric density; m is the vehicle mass; S is the reference area; μ_M is the gravitational constant of the earth; w_e is the earth self-rotation rate. The differentiation is with respect to time t . C_D and C_L are the drag and lift coefficients, respectively. These coefficients contain one control variable angle of attack α , whereas the other control variable is the bank angle ν .

For the entry guidance of a vehicle with a high lift-to-drag ratio, the inequality path constraints include the heating rate, load factor, and dynamic pressure:

$$Q_s = k_h \rho^{1/2} V^{3.15} < Q_{s \max}, \quad (7)$$

$$N = \sqrt{D^2 + L^2} / (mg_0) < N_{\max}, \quad (8)$$

$$q = \rho V^2 / 2 < q_{\max}, \quad (9)$$

where Q_s is the 升温速率 (in W/m²); N is the total load factor; q is the dynamic pressure; k_h is a constant coefficient; D and L are the aerodynamic drag and lift, respectively, i.e., $D = C_D q S$ and $L = C_L q S$.

The entry guidance problem can be described as follows: Given the initial condition of the vehicle, guide the vehicle to the target location $(h_f^*, \lambda_f^*, \varphi_f^*)$ under the path constraints, while satisfying the constraint of terminal velocity magnitude and eliminating the heading error. It can be expressed as follows: 消除航向

$$h_f = h_f^*, \lambda_f = \lambda_f^*, \varphi_f = \varphi_f^*, \quad (10)$$

$$V_f = V_f^*, \quad (11)$$

$$\Delta \sigma_f \leq \Delta \sigma_{\max}, \quad (12)$$

where the terminal time t_f is not fixed, and $\Delta \sigma$ denotes the heading error with $\Delta \sigma_{\max}$ as a pre-specified tolerance.

The guidance problem involves determining the angle of attack profile and bank angle profile.

For the high lift-to-drag ratio vehicles with low orbital velocities investigated in this paper, the Coriolis inertial force and centrifugal inertial force caused by the earth's self-rotation are small, compared with the aerodynamic force and gravity, whose accelerations are approximately $10^{-2}g_0$ and $10^{-3}g_0$, respectively [16]. Therefore, the non-rotating spherical earth assumption is employed in the analytical guidance law design. The equations of motion in a vertical plane can then be simplified as follows:

$$\begin{cases} \dot{V} = -C_D \rho V^2 S / (2m) - g \sin \theta, \\ \dot{\theta} = C_L \rho V S \cos \nu / (2m) - g \cos \theta / V \\ \quad + V \cos \theta / (R_e + h), \\ \dot{h} = V \sin \theta, \\ \dot{L}_R = R_e V \cos \theta / (R_e + h), \end{cases} \quad (13)$$

where $h = r - R_e$ is the flight altitude; L_R is the range; $g = \mu_M / r^2$ is the gravity acceleration.

2.2 Quasi-equilibrium glide condition modeling

The equilibrium glide concept was first introduced during the analysis of a hypervelocity lifting flight by the German rocket scientist Sänger as early as the 1930s. In hypervelocity gliding, a balance is maintained between the lifting force

and the gravity, which enables the vehicle to glide gently. This makes it possible to avoid trajectory oscillation. When the bank angle is changeable, it is termed as quasi-equilibrium glide. Quasi-equilibrium glide has two definitions. One assumes that the flight path angle is constant, i.e., $\dot{\theta} = 0$, and the other assumes that the descent rate is constant, i.e., $\dot{h} = 0$. The former is used in traditional quasi-equilibrium glide guidance method [8], while the latter is employed in the APC guidance algorithm for Mars exploration [2–5].

1) Constant flight path angle.

In this method, a constant flight path angle θ is maintained. Taking $\dot{\theta} = 0$ in the second equation in eq. (13) leads to the quasi-equilibrium glide condition (QEGC):

$$m \left(g - \frac{V^2}{r} \right) \cos \theta - L \cos \nu = 0. \quad (14)$$

2) Constant descent rate.

In order to maintain a constant altitude rate, we simply set

$$\dot{h} = 0. \quad (15)$$

By performing differentiation on the third equation in eq. (13) and combining it with the first and second equations in eq. (13), we obtain

$$\dot{V} \sin \theta + V \cos \theta \dot{\theta} = 0. \quad (16)$$

It can be seen from eq. (16) that even though $\dot{\theta} = 0$ is achieved, the condition $\theta = 0$ still needs to be satisfied. From another point of view, since $\dot{h} = V \sin \theta$ remains constant and velocity V decreases continually, the magnitude of the flight path angle θ should increase continually. Therefore, there is a definite difference between these two forms of the QEGC. Once the descent rate remains constant, the flight path angle will no longer remain constant. For the circumstances where QEGC is used in this paper, the form with the constant flight path angle in eq. (14) is adopted.

3 Initial descent guidance

Considering the possibility that QEGC may not be valid for several entry vehicles in the initial phase because of a low atmospheric density, the initial descent phase is introduced. The goal is to guide the vehicle within the path constraints for a situation where QEGC is not valid, and then continuously evaluate the equilibrium glide ability to ensure that it switches to the quasi-equilibrium phase in time.

The inequality path constraints include those on the heating rate, load factor, and dynamic pressure, which together make up of the flight corridor. It is widely recognized that the heating rate constraint is the main factor in

the initial entry phase for high lift vehicles. Therefore, a large constant angle of attack is adopted. A constant bank angle ν_0 is used to numerically integrate the 3DOF equations of motion. It is evaluated continuously whether to switch to the quasi-equilibrium phase. A small value of the bank angle lowers the heating rate constraint and slows down the decrease in flight altitude. Thus, we first set $\nu_0 = 0$. Because a small bank angle can cause oscillations in the altitude, the iteration method is adopted to determine the bank angle. Fixed increments are applied to the current value of ν_0 until the condition for switching to quasi-equilibrium is achieved. In order to ensure a smooth transition to the QEGC profile, a switch condition similar to that in ref. [8] is adopted:

$$\left| \frac{dh}{dV} - \left(\frac{dh}{dV} \right)_{\text{QEGC}} \right| < \delta, \quad (17)$$

where $\delta > 0$ is a small pre-selected value. From the equations of motion in eq. (13), we can obtain

$$\frac{dh}{dV} = \frac{V \sin \theta}{-\frac{D}{m} - g \sin \theta}. \quad (18)$$

$(dh/dV)_{\text{QEGC}}$ is the slope of the QEGC at the current point (h, V) , which can be obtained by treating h as a function of V from the QEGC, and taking the derivative of h with respect to V :

$$\left(\frac{dh}{dV} \right)_{\text{QEGC}} = \frac{V h_s [2m \cos \theta + C_L \rho S (R_e + h) \cos \nu]^2}{mg \cos \theta [2m h_s \cos \theta + C_L \rho S (R_e + h)^2 \cos \nu]}. \quad (19)$$

In the above equation, the atmospheric density is assumed to be $\rho = \rho_0 e^{-h/h_s}$, and $h_s \approx 7.11$ km is the density scale altitude.

In the initial phase, the QEGC switch condition is judged continuously until a successful switch to the quasi-equilibrium phase is achieved. Considering the fact that a very large integration step or a very small δ value may make the switch condition difficult to satisfy, it is necessary to set the compulsive stop condition. For example, once the flight altitude is lower than $H_{\text{down}} = 55$ km, the bank angle is increased and another iteration is carried out. If the QEGC switch condition cannot be satisfied even for the pre-selected maximum bank angle ν_{max} , the normal angle of attack profile should be adjusted. Once the existing bank angle is determined via iteration, it should be maintained until the next guidance cycle. The bank angle sign is determined by performing periodic bank reversal based on a heading error corridor [1]. The QEGC switch condition is

evaluated in each guidance cycle, and the quasi-equilibrium phase begins after the switch condition is achieved.

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4 Quasi-equilibrium adaptive glide guidance

In the traditional quasi-equilibrium glide guidance [8], the angle of attack-versus-velocity profile is assumed to be given beforehand, and only the bank angle is adjusted during guidance. The quasi-equilibrium glide adaptive guidance method proposed in this paper is quite different from the conventional methods. The bank angle is primarily determined in order to control the velocity and eliminate the heading error. The angle of attack is then adjusted according to the prediction of the terminal state based on QEGC. The principle of the adaptive quasi-equilibrium guidance algorithm is shown in Figure 1.

The guidance process is implemented based on QEGC. On the one hand, the terminal velocity can be predicted based on the specific trajectory form under QEGC. Accordingly, the velocity can be controlled by adjusting the bank angle and the heading error can be eliminated. On the other hand, the analytical relationship between the flight path angle and range can be determined with the aid of QEGC. Based on this, the required reference flight path angle can be derived according to the requirement of the current range-to-go. The quasi-equilibrium glide with a specific flight path angle can be controlled by adjusting the angle of attack. In addition, the path constraint problem is converted into an angle of attack constraint problem with the aid of QEGC. The control variables, bank angle, and angle of attack that will maintain the vehicle's quasi-equilibrium glide under the path constraints are determined successively. The constraints on both the terminal position and velocity are satisfied.

4.1 Analytical predictor of terminal velocity

Regardless of the computational cost, it is easy to calculate the terminal velocity with high precision by using numerical integration. Actually, it is difficult to implement on-board applications considering the current capability of computers. With this in mind, this paper adopts an analytical approach to predict the terminal velocity. The main idea is to build an

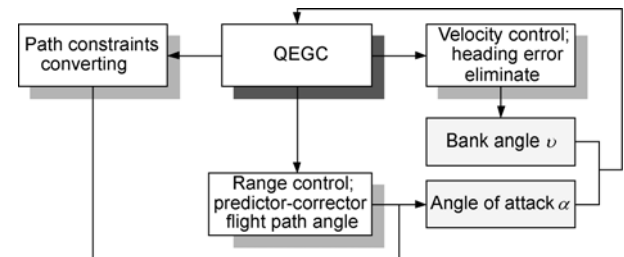


Figure 1 Principle of quasi-equilibrium adaptive guidance algorithm.

倾斜角度符号是通过基于航向误差走廊[1]执行周期性倾斜反转来确定的

analytical relationship between the terminal velocity and flight altitude by separating the effects of the drag deceleration and gravity from the velocity calculation.

In order to predict the velocity loss, only the aerodynamic drag is considered. The first equation in eq. (13) can be rewritten as

$$\frac{dV_D}{dt} = -\frac{D}{m} = -\frac{V_D^2 SC_D}{2m} \rho_0 e^{-h/h_0}, \quad (20)$$

where V_D is the velocity to be predicted when only the drag is taken into account. By combining this equation with the third equation in eq. (13), the relationship between the velocity and altitude can be obtained as

$$\frac{1}{V_D} dV_D = -\frac{C_D S \rho_0}{2m \sin \theta} e^{-h/h_0} dh. \quad (21)$$

Based on the QEGC, the flight path angle θ is a constant. Assuming that the aerodynamic drag coefficient C_D is nearly constant, the differential eq. (21) can be analytically integrated. Consequently, the terminal velocity, in relation to the aerodynamic drag, can be deduced as

$$V_{Df} = V_{\text{now}} e^{\left[h_0 \frac{C_D S \rho_0}{2m \sin \theta} e^{-h/h_0} \right]_{h_{\text{now}}}^{h_f}}, \quad (22)$$

where V_{Df} is the terminal velocity when only the drag effect is considered; V_{now} and h_{now} are the current velocity and altitude, respectively. Therefore, the velocity loss caused by aerodynamic drag can be expressed as

$$\Delta V_{\text{aero}} = V_{\text{now}} - V_{Df}. \quad (23)$$

Furthermore, the resulting terminal velocity V_f can be calculated using Keplerian laws when the orbital dynamics of a vehicle under gravity are taken into account.

$$V_f = \sqrt{V_{\text{now}}^2 + 2\mu_M \left(\frac{1}{R_e + h_f} - \frac{1}{R_e + h_{\text{now}}} \right)} + \Delta V_{\text{aero}}. \quad (24)$$

Given the vehicle parameters and flight altitudes of the initial state and terminal state, the terminal velocity of the vehicle based on QEGC can be calculated using eq. (24). It should be emphasized that the velocity prediction is implemented in each guidance cycle, and there may be small prediction errors in the initial phase due to small fluctuations in the aerodynamic coefficients and flight path angle. However, the prediction precision can gradually be improved as the vehicle approaches the target point. The terminal prediction value is accurate. The closed-loop guidance characteristics of the proposed algorithm will gradually eliminate the prediction error.

航向误差消除和速度控制

4.2 Heading error elimination and velocity control

In this paper, both heading error elimination and velocity

航向误差定义为视线与速度方向之间的角度 (可由速度方位角表示):

control are implemented by adjusting the bank angle. The heading error is defined as the angle between the line-of-sight and the velocity directions (which can be indicated by the velocity azimuth angle):

$$\Delta \sigma = \sigma_{\text{LOS}} - \sigma, \quad (25)$$

where σ_{LOS} is the line-of-sight angle, which can be calculated using the spherical triangle theory. In order to eliminate the heading error, the bank angle is adjusted by using proportional error feedback:

$$\nu_1 = |k_\sigma \Delta \sigma|, \quad (26)$$

where ν_1 is the bank angle for eliminating the heading error, and k_σ is the feedback gain. Obviously, when there is no heading error, there is no need to implement bank reversal from left to right, which can save energy.

In order to realize velocity control, on the basis of the bank angle used for eliminating the heading error, an additional bank angle is added to correct the terminal velocity. It is not difficult to discover the relationship between the bank angle and terminal velocity: the larger the bank angle is, the faster the altitude decreases. The velocity loss caused by aerodynamic drag becomes larger with the increasing of atmospheric density, resulting in a smaller terminal velocity for a certain range. Therefore, the additional bank angle can be calculated using terminal velocity error feedback as follows:

$$\delta \nu = \begin{cases} k_v (V_{f_predict} - V_f^*), & V_{f_predict} > V_f^*, \\ 0, & V_{f_predict} \leq V_f^*, \end{cases} \quad (27)$$

where $V_{f_predict}$ is the predicted terminal velocity; V_f^* is the expected terminal velocity; k_v denotes the positive error feedback gain.

Finally, the resulting bank angle magnitude is calculated as

$$|\nu| = \nu_1 + \delta \nu. \quad (28)$$

The heading error feedback gain and terminal velocity error feedback gain should be selected synthetically. However, the performance is sensitive to neither of these two coefficients. Once they are determined, adjustments due to change in guidance mission are not required. The bank angle sign is determined by performing periodic bank reversal based on a heading error corridor [1].

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4.3 Converting path constraints 转换路径约束

A major shortcoming in the existing predictor-corrector algorithms is the inability to enforce 'hard' inequality path constraints. To enforce inequality path constraints in the guidance algorithm directly, a novel algorithm is proposed

将路径约束转换为攻角约束
to convert the path constraints into angle of attack constraints. The atmospheric density is expressed as $\rho = \rho_0 e^{-h/h_0}$, and the aerodynamic drag and lift are expressed as $D = C_D \rho V^2 S/2 = C_D \rho_0 e^{-h/h_0} V^2 S/2$ and $L = C_L \rho V^2 S/2 = C_L \rho_0 e^{-h/h_0} V^2 S/2$, respectively.

For the total load factor constraint, $N = \sqrt{D^2 + L^2} / mg_0 < N_{\max}$, the functional dependency can be expressed as $N = F_N(V, h, \alpha)$. Given the current states V and h of the vehicle, the upper bound of the corresponding angle of attack $\alpha_{\max-N}$ can be determined using the equation $F_N(V, h, \alpha) = N_{\max}$. From a comparison between $\alpha_{\max-N}$ and the limit angle of attack, the smaller one is selected as the upper bound of the angle of attack, denoted by α_{up} .

The heating rate constraint, $Q_s = k_h \rho^{1/2} V^{3.15} < Q_{s\max}$, and dynamic pressure constraint, $q = \rho V^2 / 2 < q_{\max}$, can be expressed as $Q_s = F_Q(h, V)$ and $q = F_q(h, V)$, respectively. Neither of the expressions for these constraints includes the angle of attack explicitly. Thus, it is difficult to directly determine the control variable constraint based on the current trajectory state. To address this problem, the QEGC is employed to realize constraint conversion. The lower boundaries of the flight corridor in altitude-velocity (H-V) space are calculated first, consisting of the heating rate and dynamic pressure constraints, which are expressed as

$$H_Q = f_Q(V) = -2 \ln \left(\frac{Q_{s\max}}{k_h \rho_0^{1/2} V^{3.15}} \right) \cdot h_s, \quad (29)$$

$$H_q = f_q(V) = -\ln \left(\frac{2q_{\max}}{\rho_0 V^2} \right) \cdot h_s. \quad (30)$$

It is obvious that both H_Q and H_q are increasing functions of velocity V . The altitude h in the QEGC expression eq. (14) is replaced by these variables, and V is obtained from the current flight state. Given bank angle ν , the lower bounds of the corresponding angle of attack $\alpha_{\min-Q}$ and $\alpha_{\min-q}$ can be calculated using eq. (14), and the larger one should be selected as the final lower bound of the angle of attack α_{down} . Therefore, the flight path constraints should be satisfied as long as the angle of attack meets the following constraint:

$$\alpha_{\text{down}}(V) \leq \alpha(V) \leq \alpha_{\text{up}}(V). \quad (31)$$

距离控制和攻角确定

4.4 Range control and angle of attack determination

By assuming that θ is a constant with a small magnitude and setting $\sin \theta = \theta$ and $\cos \theta = 1$, the third and forth equations in eq. (13) can be used to obtain

$$dL_R = \frac{R_e}{\theta} \frac{1}{R_e + h} dh. \quad (32)$$

By integrating dL_R , we can obtain the range from the initial altitude to the current altitude, expressed as

$$L_R = \frac{R_e}{\theta} \ln(R_e + h) \Big|_{h_1}^h = \frac{R_e}{\theta} \ln \frac{R_e + h}{R_e + h_1}, \quad (33)$$

where h_1 and h are the initial and current altitudes, respectively.

Based on the above analytical relationship between the flight path angle and range, the required flight path can be determined in real-time according to the required range. The trajectory with the required flight path angle is controlled by adjusting the angle of attack. The chief steps are as follows: 1) Set the initial value of flight path angle θ . The terminal range L_R can be predicted according to the analytical relationship, i.e., eq. (33). 2) Correct the flight path angle based on the error between the predicted range and the required range until the required range is achieved. The required flight path angle θ_{need} can then be obtained. 3) Maintain the required flight path angle using trajectory control within this guidance cycle.

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Given the range-to-go, the required flight path angle can be determined using the analytical eq. (33).

$$\theta_{\text{need}} = R_e \ln \frac{R_e + h}{R_e + h_1} / L_{R_{\text{logo}}}, \quad (34)$$

where $L_{R_{\text{logo}}}$ is the current range-to-go, calculated by using the spherical triangle theory given the current longitude-latitude (λ, φ) and the target longitude-latitude $(\lambda_f^*, \varphi_f^*)$. θ_{need} is the current required flight path angle. Obviously, the requests to realize quasi-equilibrium glide adaptive guidance in this guidance cycle are

$$\begin{cases} \dot{\theta} = 0, \\ \theta = \theta_{\text{need}}. \end{cases} \quad (35)$$

For the above trajectory control problem, we first adjust the flight path angle to θ_{need} , and maintain it. A proportional guidance law is introduced to solve the problem, i.e.,

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$$n_y^* = -K(\theta - \theta_{\text{need}}) + g / g_0 \cdot \cos \theta_{\text{need}}, \quad (36)$$

where K is a pre-selected feedback gain, and $K > 0$. Note that the bank angle magnitude ν is given, and that the angle of attack can be obtained using the required load factor n_y^* .

Once the flight path angle is equal to θ_{need} , it needs to be maintained. Substituting the QEGC (eq. (14)) into eq. (13) yields

$$\frac{C_L \rho V S}{2m} \cos \nu + \left(\frac{V}{R_e + h} - \frac{g}{V} \right) \cos \theta_{\text{need}} = 0. \quad (37)$$

Thereby, the angle of attack $\alpha(V)$ can be obtained via aerodynamic parameter interpolation. The resulting angle of attack, taking into account the path constraints, is

$$\alpha = \begin{cases} \alpha_{\text{down}}(V), & \text{if } \alpha(V) < \alpha_{\text{down}}(V), \\ \alpha(V), & \text{if } \alpha_{\text{down}}(V) \leq \alpha(V) \leq \alpha_{\text{up}}(V), \\ \alpha_{\text{up}}(V), & \text{if } \alpha(V) > \alpha_{\text{up}}(V). \end{cases} \quad (38)$$

In the guidance process, the range-to-go is analytically predicted using eq. (33) with the specified terminal altitude as the stopping condition. Thus, the required flight path angle can be obtained. The trajectory control is implemented to maintain the quasi-equilibrium glide with respect to the required flight path angle by adjusting the angle of attack. In order to reduce the implementation burden of the attitude control system, the varying rate of the guidance commands should be restricted. Take the angle of attack as an example. It is assumed that the angle of attack in the current time is α_{now} , the angle of attack command without a rotation rate limit in the next time is α_{cmd} , and the interval (i.e., guidance cycle) is Δt . Then, the guidance command with a rotation rate limit is

$$\alpha_{\text{next}} = \begin{cases} \alpha_{\text{now}} + |\dot{\alpha}|_{\text{max}} \Delta t, & \alpha_{\text{cmd}} - \alpha_{\text{now}} > |\dot{\alpha}|_{\text{max}} \Delta t, \\ \alpha_{\text{cmd}}, & -|\dot{\alpha}|_{\text{max}} \Delta t \leq \alpha_{\text{cmd}} - \alpha_{\text{now}} \leq |\dot{\alpha}|_{\text{max}} \Delta t, \\ \alpha_{\text{now}} - |\dot{\alpha}|_{\text{max}} \Delta t, & \alpha_{\text{cmd}} - \alpha_{\text{now}} < -|\dot{\alpha}|_{\text{max}} \Delta t. \end{cases} \quad (39)$$

The rotation rate limit of the bank angle is similar to that of the angle of attack.

If the rotation rate is too large, there will be some guidance errors caused by the angle rate limit. However, since the proposed algorithm belongs to the group of closed-loop guidance methods, and rotation rate saturation cannot appear all the time, the errors will gradually be eliminated. Actually, the guidance law belongs to the analytical predictor-corrector (APC) guidance methods under the QEGC condition. It is independent of the reference trajectory, and all the guidance commands are calculated in real time.

5 Testing the guidance algorithm

5.1 Vehicle and guidance mission

Lockheed-Martin's CAV-H vehicle was employed in the simulation. It is a typical hypersonic high lifting entry vehicle [17]. The vehicle weight is 907 kg, the reference area is 0.4839 m^2 , and the maximum lift-to-drag ratio $(L/D)_{\text{max}}$ =

3.5. The simulation was carried out over an oblate rotating earth, and the 1976 US standard atmosphere model was adopted.

The path constraints on the entry trajectories are as follows:

- 1) The peak heating rate constraint, $Q_{\text{max}} = 2 \text{ MW/m}^2$.
- 2) The peak load factor constraint, $N_{\text{max}} = 4$.
- 3) The peak dynamic pressure constraint, $q_{\text{max}} = 200 \text{ kPa}$.

The vehicle flight control authority is restricted by the following conditions.

- 1) Assuming no sideslip and a bank-to-turn (BTT-180) turn mode, the angle of attack always has a positive value.
- 2) The maximum angle of attack is $\alpha_{\text{max}} = 20^\circ$, and the maximum bank angle is $|\nu|_{\text{max}} = 70^\circ$.
- 3) The maximum rates are $|\dot{\alpha}|_{\text{max}} = 10^\circ/\text{s}$ and $|\dot{\nu}|_{\text{max}} = 20^\circ/\text{s}$.

The guidance mission is as follows. Given the initial state of the vehicle, lead the vehicle smoothly glide to the target point such that all the path and terminal constraints are satisfied, including those for the terminal longitude, latitude, altitude, velocity magnitude, and heading angle.

5.2 Nominal flight performance

The initial settings of the simulation parameters are altitude $h_0 = 70 \text{ km}$, velocity $V_0 = 6500 \text{ m/s}$, and flight path angle $\theta_0 = 0$. The longitude and latitude are both 0° . The expected target longitude and latitude are 120° and 40° , respectively. The terminal altitude is 25 km , and the terminal velocity is 2400 m/s .

The main results are shown in Figure 2. Figure 2(a)–(d) show that the trajectory is very smooth, except for the obvious switches in the altitude and flight path angle during the period of switching from the initial phase to the quasi-equilibrium glide phase. These results indicate the advantage of the quasi-equilibrium glide adaptive guidance method. As can be seen, the profile of the flight path angle is very close to zero, except during the initial descent phase. It implies that the QEGC is well satisfied. Figure 2(e) and (f) show that the control variables α and ν are generally smooth. For the angle of attack, only a few small magnitude oscillations occur during the bank reversal. Because the steering angle rate limits have been considered in the guidance method, the steering system is not excessively burdened. Thus, the system can be used in practice. Since the altitude is treated as the stopping condition in simulation, the terminal altitude constraint is met accurately. The terminal longitude and latitude are 120.997° and 40.002° , respectively, and the terminal position deviation is approximately 500 m , indicating a high guidance precision. The terminal velocity deviation is -27 m/s . The bank reversal and heading error convergence processes are shown in

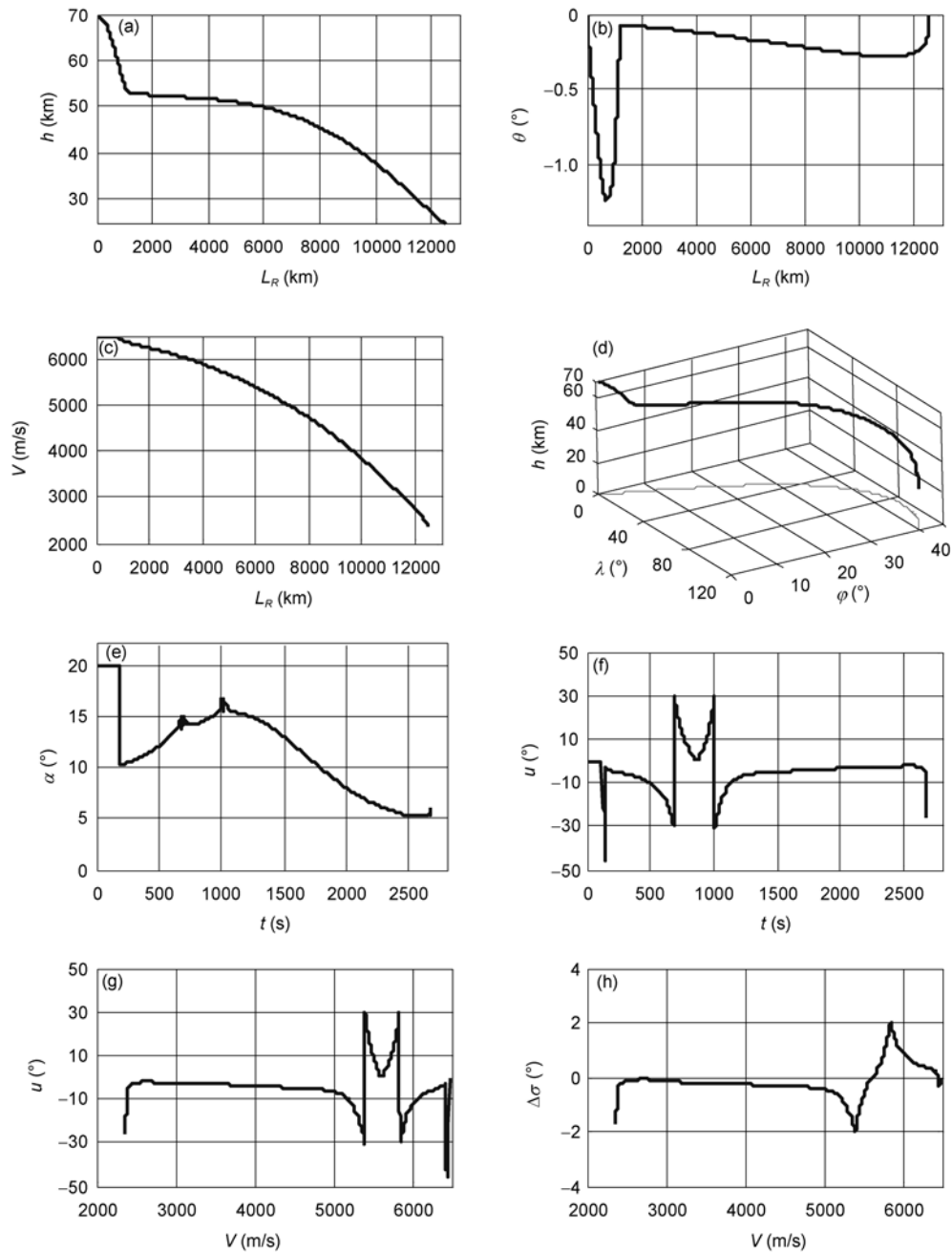


Figure 2 Simulation results for nominal performance. (a) Altitude-range; (b) flight path angle-range; (c) velocity-range; (d) altitude-longitude-latitude; (e) angle of attack-time; (f) bank angle-time; (g) bank angle-velocity; (h) heading error-velocity.

Figure 2(g) and (h), respectively. As can be seen, there are two bank reversals at the boundary of the heading error corridor, and the terminal heading error is approximately 1.7° . The simulation results clearly indicate that all the terminal constraints are satisfied.

The path constraints in both the drag-velocity (D-V) profile and angle of attack profile are satisfied, as illustrated in Figure 3. The path constraints described in the D-V profile have been successfully used in shuttle entry guidance [1]. Accordingly, the angle of attack profile with the path constraints investigated in this paper shows good conversion

performance for the path constraints. This indicates that, in the D-V profile, the intersection of the heating rate, load factor, and dynamic pressure is the upper bound, while the quasi-equilibrium glide is the lower bound. In the constrained angle of attack profile, the heating rate and dynamic pressure are part of the lower bound, while the load factor is part of the upper bound. It can be seen from the D-V profile that the generated vehicle trajectory profile is almost superimposed on the quasi-equilibrium glide boundary. This proves that the trajectory generated by the proposed guidance algorithm can suitably satisfy the QEGC.

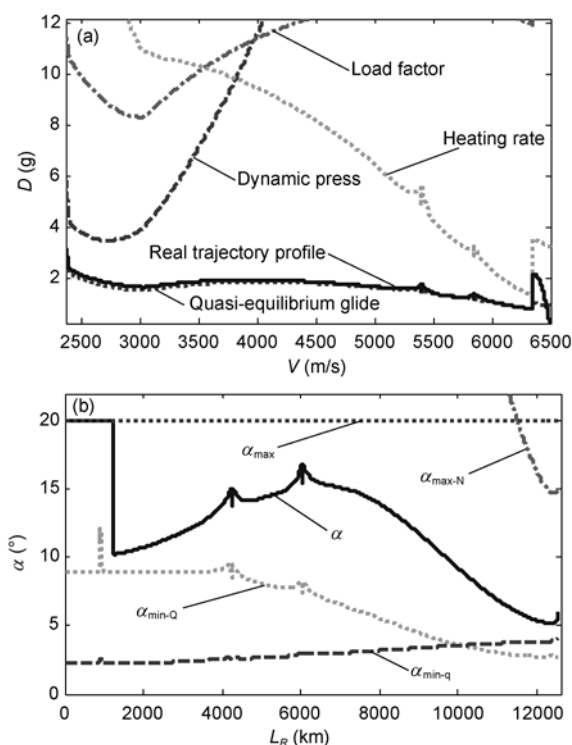


Figure 3 Path constraint conversion performance. (a) Drag-velocity (D-V) flight corridor; (b) converted path constraints expressed by angle of attack.

Moreover, there is another advantage that all the path constraints are easy to satisfy during most of the flight time.

5.3 Adaptive performance for mission change

Any deviation from the “hard” path constraint will cause the entire mission to fail. In order to evaluate the adaptability of the proposed guidance algorithm with respect to the path constraints, the constraints were artificially tightened in the simulation. As an example, the dynamic pressure constraint was tightened from $q_{max}=200$ kPa to $q_{max}=170$ kPa. At the same time, the required terminal velocity was adjusted to 3000 m/s in order to facilitate a comparison.

A comparison between the angle of attack profiles with and without modulation for the path constraint is shown in Figure 4. The angle of attack without modulation is denoted by the thick dashed line. It exceeds the dynamic pressure constraint towards the end, whereas the angle of attack with modulation, denoted by the thick solid line, remains at the dynamic pressure constraint boundary even under the most unfavorable conditions. Therefore, the proposed constraint conversion method is verified to be adaptive and highly feasible.

The adaptive guidance concept proposed in this paper is that the guidance algorithm is able to adapt to a new guidance mission without adjusting the guidance parameters within the vehicle's flight ability. One of the most important

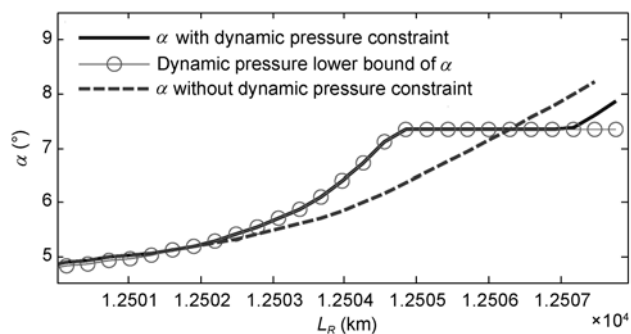


Figure 4 Angle of attack adaptive modulation under path constraints (using dynamic pressure as example).

characteristics lies in its adaptability to different target points, as well as different required terminal velocities. In order to evaluate the adaptive performance for a target point change, we set several different target points under the same initial condition, including ($\lambda_f=100^\circ$, $\varphi_f=10^\circ-80^\circ$) and ($\lambda_f=10^\circ-90^\circ$, $\varphi_f=80^\circ$), and the results are shown in Figure 5. It can be seen that when the maximum movement scope of the target point is more than 1000 km, the proposed algorithm can automatically lead the vehicle to any target point without adjusting the guidance parameters and all the terminal location deviations are less than 1 km. In order to obtain the adaptive performance for the required terminal velocity change, the target point was fixed at ($\lambda_f=120^\circ$, $\varphi_f=40^\circ$), and different required terminal velocities (from 2000 to 3000 m/s) were selected. As shown in Figure 6, all the terminal velocity errors are within 50 m/s. The above simulation results demonstrate the adaptability of the proposed method, which enables the vehicle to have high robustness to the initial deviation and provides considerable convenience in the case of mission change.

5.4 Monte Carlo simulation

In order to test the guidance performance and robustness

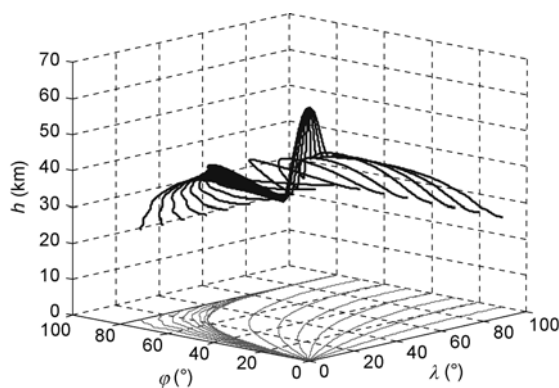


Figure 5 Trajectory profiles for different targets.

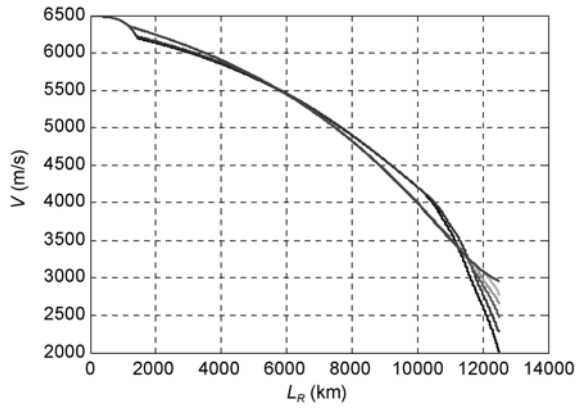


Figure 6 Velocity profiles for different required terminal velocities.

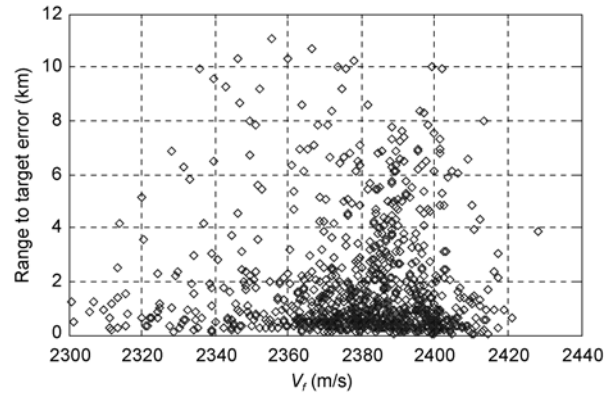


Figure 8 Range-to-target errors and terminal velocities.

under a wide range of common random dispersions encountered during an entry flight, Monte Carlo 3DOF simulations were conducted. Since the robustness of the algorithm under the initial dispersions was validated as described in the previous section, this part will focus on testing the performance with respect to the aerodynamic coefficient and atmospheric density dispersions. Gaussian random dispersions with $3\sigma=15\%$ were added to the aerodynamic coefficients and atmospheric density. It was assumed that the guidance system has no knowledge of these dispersions. Figure 7 illustrates the dispersions of the terminal points for 1000 dispersed trajectories, and Figure 8 shows the range-to-target errors and terminal velocities. The figures indicate that the terminal position circular error probability (CEP) is less than 2 km and that the terminal velocity error is less than 40 m/s. It proves that the guidance algorithm has a strong robustness to the dispersions.

6 Conclusion

This study presents a novel quasi-equilibrium glide adaptive guidance algorithm for hypersonic atmospheric entry vehicles. The main characteristic of the proposed methodology

lies in the fact that it takes full advantage of the QEGC to generate an excellent quasi-equilibrium glide with a smooth trajectory. At the same time, precise predictions of range and velocity, which are crucial for analytically calculating the guidance commands, are realized through the use of the QEGC. In addition, the convenient conversion of the path constraints into the upper and lower bounds of the angle of attack is also presented. The major advantages of the proposed algorithm are as follows: 1) It functions independently of the standard trajectory, and can automatically lead the vehicle to any target or any terminal velocity without requiring guidance parameters to be adjusted. This provides considerable convenience for an occasional change in the mission. 2) The trajectory is quite smooth due to the use of the QEGC. 3) All the path constraints can be converted into angle of attack constraints. The path constraints thus become relatively simpler, and are easily satisfied since the trajectory generated by the guidance law operates along the quasi-equilibrium glide boundary most of the time. 4) All the guidance commands are calculated analytically in real time, making it convenient for practical use.

The main drawback of the proposed guidance algorithm may lie in the possible oscillation of the steering angles. It is possible to be solved by introducing a damper into the steering equation, as a part of further research. In general, the proposed algorithm can provide some reference for the entry guidance of an RLV, hypersonic vehicle, etc.

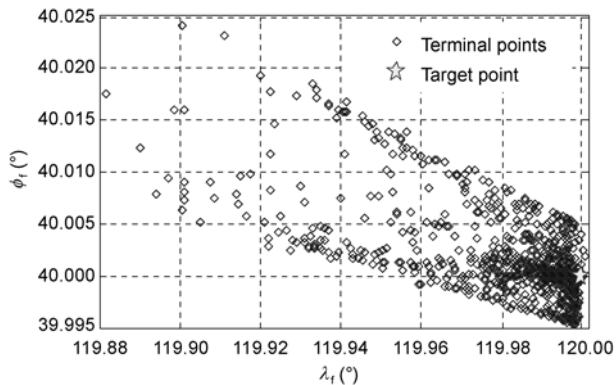


Figure 7 Longitudes and latitudes of terminal points.

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